Giulio Arecco

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LinkedIn ↗ | GitHub ↗ | Portfolio ↗

Education

Politecnico di Torino, MS in Computer Engineering — Sept 2023 – March 2026

- Specialization: Computer Graphics and Multimedia
- Final Mark: 110/110
- Master's Thesis: "Entertainment games for complex problem-solving" (Thesis PDF ↗)
- Relevant Coursework: Computer Graphics, Game Design, Virtual Reality, Computer Animation, System and Device Programming

Politecnico di Torino, BS in Computer Engineering — Sept 2020 – March 2024

Selected Projects

Beyond The Line – Master's Thesis Serious Game — GitHub ↗ | Itch.io ↗

- Developed a text-based serious game featuring dynamic narrative branching, designed to evaluate players' Complex Problem Solving skills through resource management and systemic uncertainty.
- Tools Used: Unity, C#, Ink

Unreal Engine 5 Action RPG Systems — GitHub ↗

- Developed a third-person Action RPG prototype combining C++ and Blueprints, featuring FSM-driven enemy AI, non-repeating patrol path algorithms, directional hit reactions, and event driven UI.
- Tools Used: Unreal Engine 5, C++, Blueprints

The Plague in Milan – 3D Interactive Experience — GitHub ↗ | Itch.io ↗

- Engineered a physics-driven first-person controller, an event-driven modular progression system, and a dialogue system to interact with non-player characters.
- Tools Used: Unity, C#

Monte Carlo Path Tracer — GitHub ↗

- Developed a dependency-free CPU path tracer featuring compile-time zero-cost polymorphism, explicit memory ownership, BVH acceleration, and custom À-Trous/Bilateral spatial denoisers.
- Tools Used: Zig

OS/161 Demand Paging — GitHub ↗

- Implemented a demand-paging kernel subsystem for the OS/161 operating system featuring on-demand lazy ELF loading, a FIFO page replacement queue, disk-backed swap management, and software TLB handling.
- Tools Used: C, OS/161

Technologies & Skills

Programming Languages: C++, C#, C, Zig

Game Engines: Unreal Engine 5, Unity

Developer Tools: Git, JetBrains Rider, Visual Studio, WSL/Linux

Spoken Languages: English (C1), Italian (Native)